Kernix 0.11.0 / Performance brief

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

35 workloads

123 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

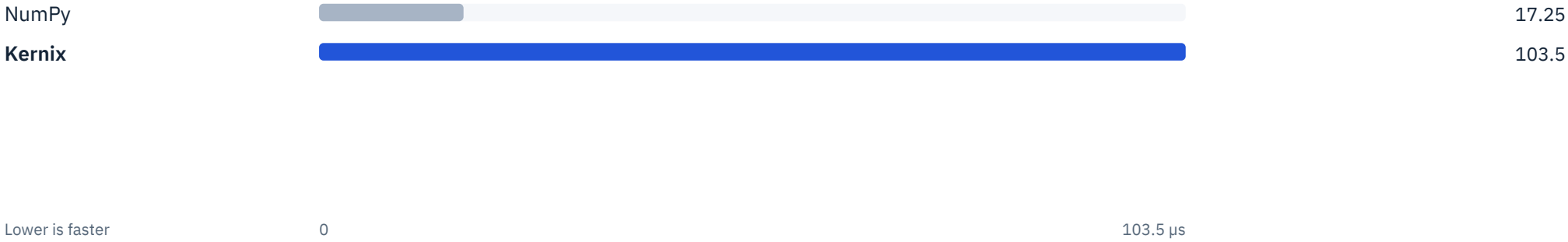
Kernix 0.11.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Independent matrix operands

Per-call median

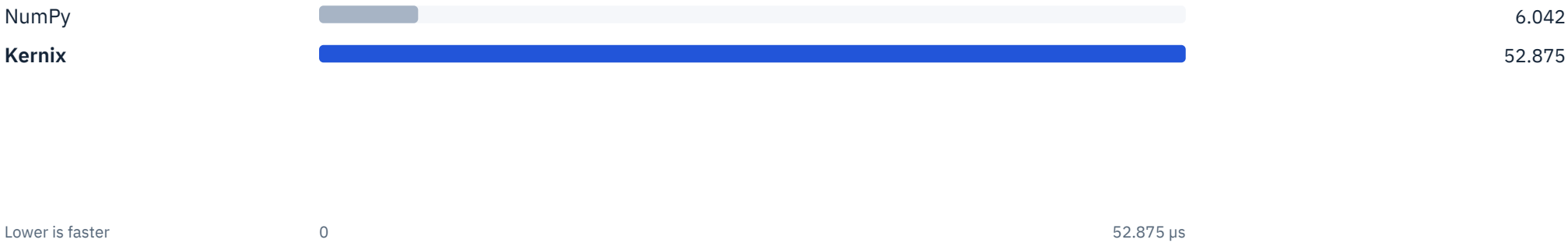
128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



Kernix 0.11.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



Kernix 0.11.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



Kernix 0.11.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



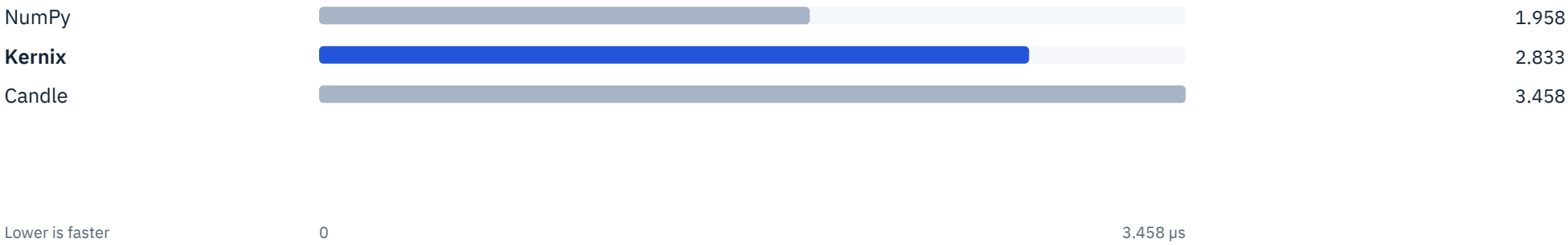
Kernix 0.11.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.11.0 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.0 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



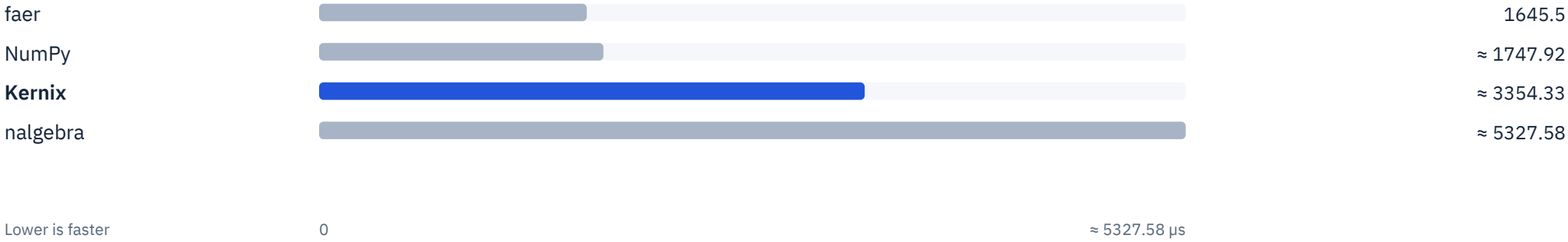
Kernix 0.11.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

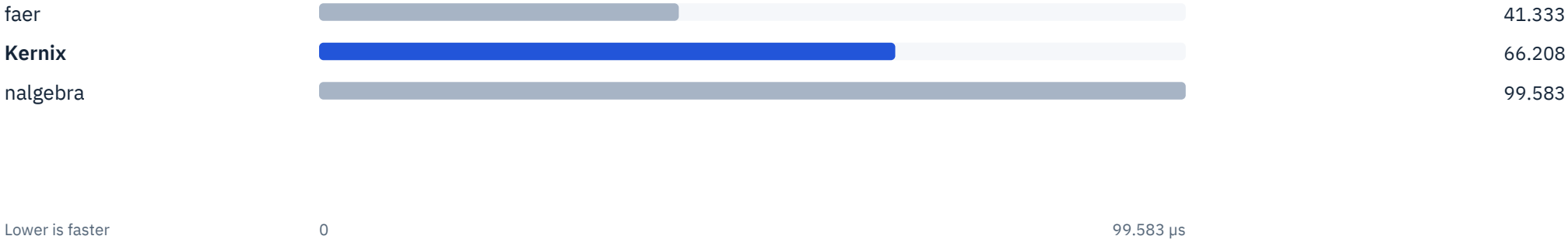
Per-call median



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



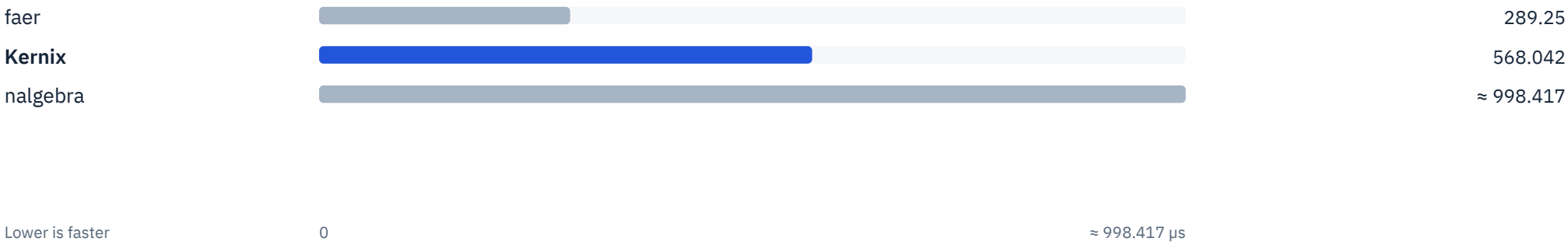
Kernix 0.11.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

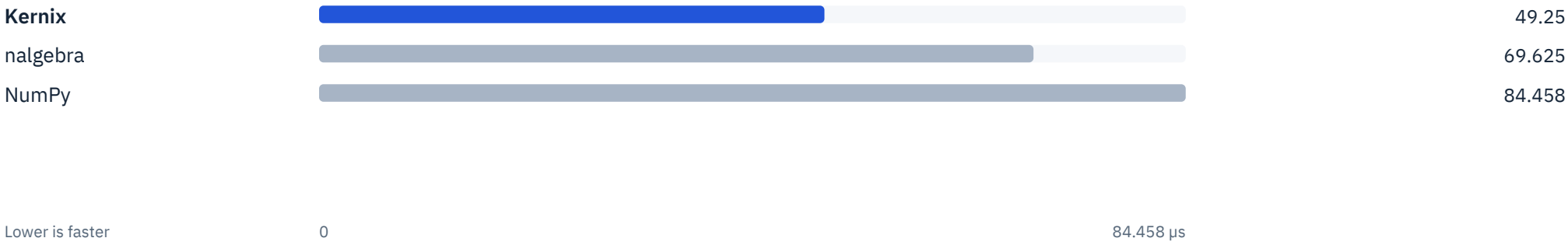
Per-call median



Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



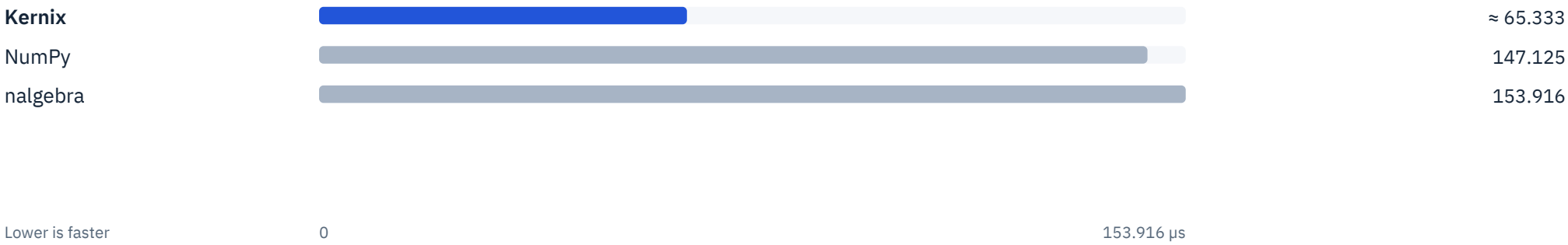
Kernix 0.11.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



Kernix 0.11.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Thin QR

256 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



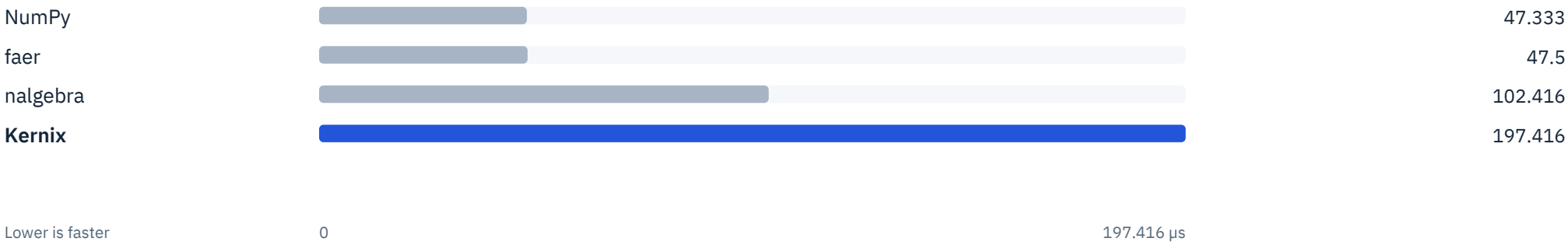
Kernix 0.11.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

Per-call median

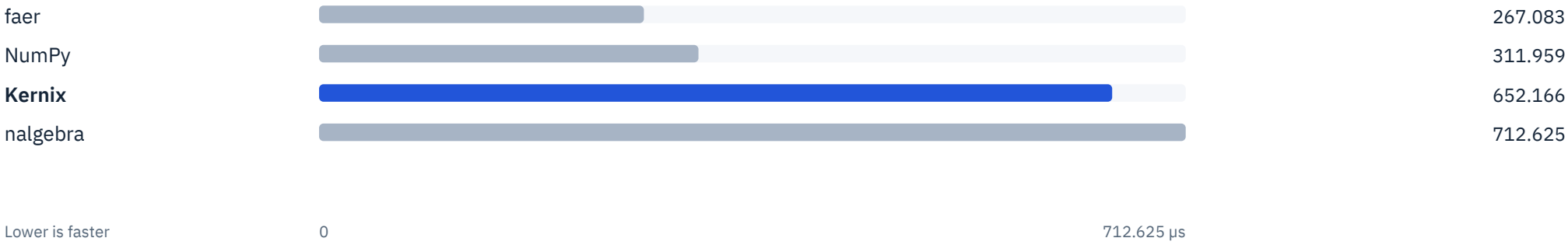
128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.11.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

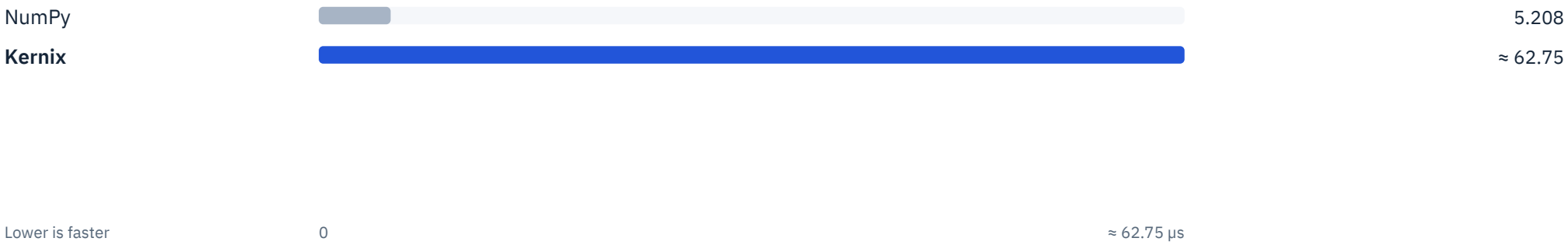


Kernix 0.11.0 / Tensor contractions

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output



Kernix 0.11.0 / Statistics & activations

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Row softmax

256 × 64 · f64 · 1 worker · allocated output



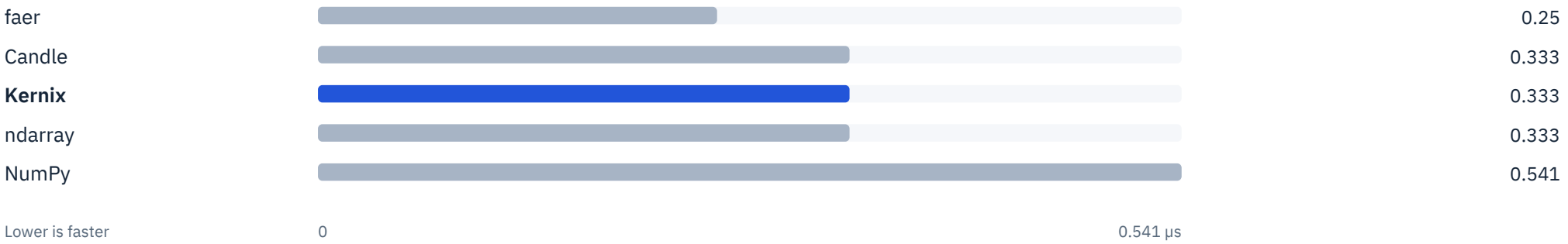
Kernix 0.11.0 / Tensor construction

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Create zeros

Per-call median

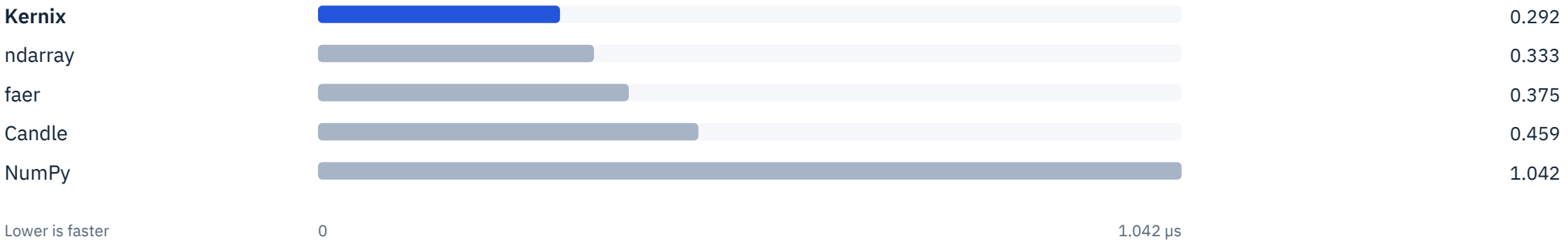
64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	17250	17.25
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	103500	103.5
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	6042	6.042
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	52875	52.875
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2083	2.083
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2041	2.041
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2625	2.625
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2417	2.417
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	1458	1.458
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2250	2.25

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	13833	13.833
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	3250	3.25
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	14708	14.708
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	14750	14.75
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	13542	13.542
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	13875	13.875
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	101708	101.708
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	16542	16.542
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	103208	103.208
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	105334	105.334

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	98292	98.292
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	98000	98
Matrix multiplication / Kernix · explicit Accelerate 128 × 128 · 128 × 128 · f64	17292	17.292
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	108167	108.167
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	792000	792
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	789083	789.083
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	812416	812.416
Matrix multiplication / Kernix · explicit Accelerate 256 × 256 · 256 × 256 · f64	121792	121.792
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	809500	809.5
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	778167	778.167

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7000	7
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	6958	6.958
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	8167	8.167
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	1916	1.916
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	7625	7.625
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	7875	7.875
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	53167	53.167
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	56708	56.708
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	5750	5.75
Matrix multiplication / Kernix · explicit Accelerate 128 × 128 · 128 × 128 · f32	5958	5.958

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	49417	49.417
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	52833	52.833
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	64292	64.292
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	32917	32.917
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	389250	389.25
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	387208	387.208
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	392417	392.417
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	412292	412.292
Matrix multiplication / Kernix · explicit Accelerate 256 × 256 · 256 × 256 · f32	34125	34.125
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	412375	412.375

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3458	3.458
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	1958	1.958
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2833	2.833
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2500	2.5
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	2209	2.209
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	1750	1.75
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	20042	20.042
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	69125	69.125
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	420083	420.083
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	45000	45

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 512 × 512 · f64	5327584	5327.584
Cholesky factorization / NumPy 512 × 512 · f64	1747916	1747.916
Cholesky factorization / faer 512 × 512 · f64	1645500	1645.5
Cholesky factorization / Kernix 512 × 512 · f64	3354333	3354.333
Column-pivoted QR / nalgebra 96 × 32 · f64	99583	99.583
Column-pivoted QR / Kernix 96 × 32 · f64	66208	66.208
Column-pivoted QR / faer 96 × 32 · f64	41333	41.333
Column-pivoted QR / faer 256 × 64 · f64	289250	289.25
Column-pivoted QR / Kernix 256 × 64 · f64	568042	568.042
Column-pivoted QR / nalgebra 256 × 64 · f64	998416.9999999999	998.4169999999999

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	69625	69.625
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	84458	84.458
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	49250	49.25
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	153916	153.916
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	147125	147.125
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	65333.00000000001	65.33300000000001
Thin QR / NumPy 96 × 32 · f64	49291	49.291
Thin QR / Kernix 96 × 32 · f64	57583	57.583
Thin QR / nalgebra 96 × 32 · f64	42625	42.625
Thin QR / faer 96 × 32 · f64	46833	46.833

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 256 × 64 · f64	304666	304.666
Thin QR / Kernix 256 × 64 · f64	522958	522.958
Thin QR / nalgebra 256 × 64 · f64	378791	378.791
Thin QR / NumPy 256 × 64 · f64	385750	385.75
Cholesky factorization / faer 64 × 64 · f64	10417	10.417
Cholesky factorization / NumPy 64 × 64 · f64	14917	14.917
Cholesky factorization / nalgebra 64 × 64 · f64	20625	20.625
Cholesky factorization / Kernix 64 × 64 · f64	22042	22.042
Cholesky factorization / NumPy 128 × 128 · f64	47333	47.333
Cholesky factorization / nalgebra 128 × 128 · f64	102416	102.416

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / faer 128 × 128 · f64	47500	47.5
Cholesky factorization / Kernix 128 × 128 · f64	197416	197.416
Cholesky factorization / Kernix 256 × 256 · f64	652166	652.166
Cholesky factorization / NumPy 256 × 256 · f64	311959	311.959
Cholesky factorization / faer 256 × 256 · f64	267083	267.083
Cholesky factorization / nalgebra 256 × 256 · f64	712625	712.625
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4082.9999999999995	4.0829999999999995
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	5792	5.792
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	6458	6.458
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3208	3.208

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model inference / Candle batched normalized/sliced MLP · f64	5375	5.375
Model inference / Kernix batched normalized/sliced MLP · f64	2375	2.375
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3333	3.333
Model recording + forward / Candle batched normalized/sliced MLP · f64	7625	7.625
Model backward / Kernix prepared graph → fresh gradients · f64	3583	3.583
Model backward / Candle prepared graph → fresh gradients · f64	20166	20.166
Parameter update / Kernix transactional SGD, fixed gradients · f64	750	0.75
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	62749.99999999999	62.74999999999999
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	5208	5.208
Row softmax / ndarray 256 × 64 · f64	61208	61.208

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Row softmax / Candle 256 × 64 · f64	67417	67.417
Row softmax / Kernix 256 × 64 · f64	79125	79.125
Row softmax / NumPy 256 × 64 · f64	74416	74.416
Create zeros / faer 64 × 64 · f32	250	0.25
Create zeros / Candle 64 × 64 · f32	333	0.333
Create zeros / NumPy 64 × 64 · f32	541	0.541
Create zeros / Kernix 64 × 64 · f32	333	0.333
Create zeros / ndarray 64 × 64 · f32	333	0.333
Create ones / ndarray 64 × 64 · f32	333	0.333
Create ones / NumPy 64 × 64 · f32	1042	1.042

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create ones / Candle 64 × 64 · f32	459	0.459
Create ones / Kernix 64 × 64 · f32	292	0.292
Create ones / faer 64 × 64 · f32	375	0.375